Firas Darwish

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Education

University of Oxford 2025 - 2029
Statistics and Machine Learning (PhD) ***Rhodes Scholar***

- Co-supervised by **Ingmar Posner** and **Yee Whye Teh**
- Efficient **Sim-and-Real** Co-training for **Generalizable Robot Manipulation** Policies
- Joint Program with Imperial College London

New York University Abu Dhabi 2021 - 2025
Bachelor of Science in Computer Engineering ***GPA: 4.000/4.000 - Summa Cum Laude***

- Supervised by **Farah Shamout** and **Mohamad Eid**
- Capstone: Building multimodal (EMG and Vision) prosthetic control algorithms.

Publications

- **Firas Darwish**, Aneesh Muppidi, Dylan Cope, Joao F. Henriques, Jakob Nicolaus Foerster. "**Finding the Time to Think: Learning Planning Budgets in Real-Time RL**." **Accepted** to **NeurIPS** 2026 (Main Track Poster).
- **Firas Darwish**, George Nicholson, Aiden Doherty, Hang Yuan. "**Motion Capture is Not the Target Domain: Scaling Synthetic Data for Learning Motion Representations**." **Accepted** to DATA-FM @ **ICLR** 2026.
- **Firas Darwish**, Dhiyaa Al Jorf, Eion Tyacke, Costanza Armanini, Farah E. Shamout, S. Farokh Atashzar. "**Leveraging Agonistic-Antagonistic Coactivation in Single-Grid HDsEMG for Hand Gesture Recognition**." **Accepted** to **EMBC** 2026.
- **Firas Darwish**, Dhiyaa Al Jorf, Eion Tyacke, Costanza Armanini, Farah E. Shamout, S. Farokh Atashzar. "**Enhancing the Efficiency of Hand Gesture Classification by Harnessing Muscle Synergies**." **Accepted** to **IEEE EMBC** 2025 (Poster Presentation).
- Leopoldo Lopez, Shaza Elsharief, Dhiyaa Al Jorf, **Firas Darwish**, Congbo Ma, Farah E. Shamout. "**Uncertainty Quantification for Machine Learning in Healthcare: A Survey**." **Accepted** to **CHIL** 2025 (Poster Presentation).
- Shaza Elsharief, Leopoldo Julian Lechuga Lopez, **Firas Darwish**, Dhiyaa Al Jorf, Mahlet Atrsaw Andargei, Farah E. Shamout. "**Modular Multimodal Alignment using Time-Series EHR Data for Enhancing Medical Image Classification**." Submitted for Review.

Research Experience

Center for Artificial Intelligence and Robotics (CAIR), NYU Abu Dhabi, UAE
Research Assistant *Feb 2024 – June 2025*

- Used **PyTorch** and **Lightning** to efficiently conduct experiments, log results, and **schedule multi-GPU and multi-node runs on NYU's High Performance Computing system** to train and evaluate neural networks.
- **MedCAM Project**: Pretrained adaptation module for enriching CXR representations with EHR data using global and contrastive loss, and tested in downstream classification and OOD detection tasks.
- **HD-sEMG Electrode Dropping Project**: Investigated redundant muscle pair activity patterns in the forearm to motivate channel dropping and reduce the computational cost of hand gesture recognition task (utilized ablation studies and GradCAM on different 3D and 4D CNN architectures).
- **Uncertainty Quantification Project**: Conducted literature review with 300+ multimodality, healthcare, and review papers scanned on Covidence.

Open-Source ML Tools Abu Dhabi, UAE
Developer *June 2024 - July 2024*

- Developed source code for use in machine learning research and projects that is made freely available to all for possible modification and redistribution (**accessed by over 200 GitHub users since release**).
- Published Python package for 3-dimensional convolutional layers using Kolmogorov-Arnold Networks (KANs)
- Published Python package with 4-dimensional implementations of MaxPool, DropBlock, and SpatialDropout.

Industry Experience

e& (Etisalat) Abu Dhabi, UAE

AI and Data Intern *May 2024 – Aug 2024*

- Lead on the **design of 2 AI use cases with UNDP** for sustainable development in the Arab World (climate resilience, service delivery) and identifying data, privacy, and security challenges.
- Defined methodology for monitoring AI impact and **instituted AI governance for 40,000 employees**.
- Developed a GenAI recommendation system on Smiles mobile app, **serving 1.6 million households in the UAE**.

WhiteBox HR Abu Dhabi, UAE

Machine Learning Intern *May 2023 – Aug 2023*

- Designed a fully automated unsupervised ML algorithm to parameterize worker well-being and performance using dimensionality reduction and clustering techniques.
- Fine-tuned an in-house generative AI chat bot as a **Natural Language Interface to DataBase (NLIDB)** used for automatically generating statistics and visualizations from tabular data.
- Deployed solutions are utilized by **30+ companies and over 2,000 employees** across the UAE and MENA region.

Competitions

- 1st **Place, eBrain Lab Generative AI Competition** - Fine-tuned foundation model to reduce model size and maximize performance on medical question-answer task.
- 2nd **Place, Mubadala Water Debris Robot Competition** — **awarded $8,200** for autonomous vehicle design.
- 3rd **place in UAE, IEEEXtreme 17.0 Competition** - 24-hour global competitive programming challenge.

Awards And Honors

- **Rhodes Scholarship**, A Partnership between the Rhodes Trust and the Salama Bint Hamdan Al Nahyan Foundation - selected as 2025 UAE Rhodes Scholar and provided full-ride graduate scholarship to Oxford University.
- **Summa Cum Laude**, NYU Abu Dhabi — awarded to **top 5%** of graduating students in Engineering.
- **Outstanding Reviewer**, DATA-FM @ ICLR 2026 — awarded to **2 reviewers** who provided exceptional reviews.
- **NYUAD Post-Graduation Practical Training Funding** — **awarded $3,200** to work on AI for Clinical Decision-Making.
- **NYUAD Research Funding** — **awarded $4,000** to work on faculty-supervised research (Summer 2023 & Summer 2024).
- **Founders Day Honors Scholar**, NYU — awarded to top-ranking baccalaureate candidates for academic excellence and scholarly distinction.
- **UAE Golden Visa for Outstanding Students** — granted for ranking among UAE's national academic toppers.